

6(a)	<u>facilitated diffusion</u> ;	1
6(b)	480 ; <i>max 1 if not as a whole number 479.5918367</i> <i>If standard form incorrect allow max 1 if evidence of 4.7 divided 9.8</i>	2
6(c)	any four from: 1 <i>ref. to</i>, many, impulses / action potentials (occurring) or many synapses / many connections between neurones ; 2. <i>ref. to</i> sodium potassium pump / active transport of Na⁺ and K⁺ ; R channel 3 setting up / maintaining, ion gradient(s) ; 4 setting up / maintaining, resting potential(s) ; 5 production / recycling, of, neurotransmitters / acetylcholine / ACh ; 6 production / recycling / movement, of vesicles or exocytosis (of neurotransmitter) / endocytosis ; R exocytosis of vesicles 7 production of acetylcholinesterase ;	4

Question	Answer	Marks
6(d)	any three from: 1 (caffeine) binds to (adenosine) receptors ; R active site or inhibits / blocks, binding of adenosine to receptors / AW ; 2 less ATP made during sleep / more ATP made whilst awake ; 3 postsynaptic (neurone) activity is not inhibited / can occur / AW ; 4 (less adenosine bound to receptors) cannot promote sleep / promotes wakefulness / AW ;	3